

MediScript AI

Integrating Findings from Related Research: A Comparative Analysis of Mediscript (IJRPR, Vol. 7, Issue 5, 2026)

Project Report Supplement — Related Work & Design Recommendations

1. Purpose of This Document

This supplement reviews a closely related published research paper, “Mediscript: A Low-Cost Multilingual Telemedicine Platform with Voice-Based Prescription Transcription and Automated NER Extraction” (International Journal of Research Publication and Reviews, Vol. 7, Issue 5, May 2026), and identifies concrete design elements from it that can strengthen MediScript AI. While the reference paper builds a doctor-facing telemedicine platform around voice dictation, and MediScript AI is a patient-facing prescription interpretation and RAG-based assistant, the two systems share a common core problem: turning unstructured medical text into safe, structured, actionable information. Six specific, transferable elements were identified and are detailed below, each mapped to an existing or new module in the MediScript AI architecture.

2. Comparative Overview

The table below summarizes how the two systems differ before detailing what is being adopted.

Dimension	Mediscript (Reference Paper)	MediScript AI (Our Project)
Primary Input	Doctor's voice dictation during live consultation	Uploaded prescription image / PDF (OCR + VLM)
Core AI Task	Speech-to-text transcription, then entity extraction	Vision-based digitization, then entity extraction
Extraction Model	Transformer encoder + BiLSTM + CRF over IOB tags	NLP pipeline (spaCy / scispaCy / Transformers) — architecture not fixed
End User	Doctor (creates the prescription)	Patient / caregiver / pharmacist (interprets the prescription)
Knowledge Grounding	None described — extraction only	RAG over verified medical vector database (FAISS / Qdrant)
Safety Model	Rule-based validation of extracted fields	Three-tier response mode + rule-based validation + confidence scoring

3. Research Insights and Their Integration

1. Structured NER Architecture (Transformer + BiLSTM + CRF)

Maps to: Module 2 — Prescription Reading (The Engine)

From the paper: Mediscript's Medication Extraction Module processes prescription text through a multilingual transformer encoder, then a BiLSTM layer for sequential context, then a CRF layer that selects the optimal IOB tag sequence, which is finally mapped to structured fields (Name, Dose, Frequency, Duration, Route).

Integration into MediScript AI:

MediScript AI's extraction step (currently OCR/VLM output passed to a general NLP pipeline) can adopt this same architecture as a defined internal stage: once OCR/VLM produces raw text, route it through a transformer-based encoder + BiLSTM + CRF sequence labeler trained on medical entities, using IOB tagging as the intermediate representation before conversion to the existing JSON schema (medicine, strength, dosage, duration, instruction). This gives the extraction module a proven, benchmarkable design instead of an unspecified “NLP pipeline,” and directly improves the Extraction Accuracy metric (Precision/Recall/F1) already listed in the evaluation plan.

- Adopt IOB tagging as the standard intermediate label format for all extracted entities.
- Use a multilingual transformer encoder so the same model spine supports Hindi, Marathi, Gujarati, and Tamil.
- Benchmark the BiLSTM+CRF head against the current spaCy/scispaCy baseline on medical entity F1 score.

2. Voice as a Secondary Input Modality

Maps to: *Module 1 — Prescription Upload & Pre-processing (new input path)*

From the paper: Mediscript's entire input pipeline is built around Whisper ASR, letting doctors dictate prescriptions in English and Hindi with noise resiliency and on-the-fly correction before parsing.

Integration into MediScript AI:

MediScript AI currently accepts only image, PDF, or camera photo input. Adding a voice input path lets a pharmacist or caregiver read an illegible prescription aloud when OCR confidence is low, or lets a patient verbally describe medicines they were told about verbally. This directly supports the existing “Pharmacists requiring quick interpretation of illegible handwriting” persona and gives the system a fallback when OCR/VLM confidence scoring flags a document as unreadable.

- Integrate Whisper (or an equivalent multilingual ASR) as an alternate entry point into Module 1.
- Route transcribed text into the same extraction pipeline as OCR output (shared downstream schema).
- Trigger the voice path automatically when AI Confidence Scoring falls below the manual-verification threshold.

3. Rule-Based Validation Layer Alongside ML Extraction

Maps to: *Module 2 output validation, tied to AI Confidence Scoring (Section 8)*

From the paper: Mediscript's extraction module includes rule-based checks that validate extracted entities against common medical standards before they are stored, in addition to the ML model's own output.

Integration into MediScript AI:

MediScript AI's planned Confidence Scoring feature currently relies on ML-derived confidence values alone. Adding a rule-based validation pass — checking that dosage units are plausible, frequency codes match a known set (OD, BD, TDS, SOS, 1-0-1, etc.), and durations fall within sane ranges — creates a second, independent safety check. A field that fails rule-based validation should force manual verification regardless of the ML confidence score, tightening the path toward the project's zero-hallucination goal.

- Define a validation ruleset for units, frequency codes, and duration ranges per medicine class.
- Combine ML confidence + rule-pass status into a single verification gate before data reaches the profile.
- Log rule violations separately from low-confidence flags to distinguish extraction errors from OCR noise.

4. Human-in-the-Loop Correction Feeding Continuous Learning

Maps to: *Module 6 (Personalized Profile) correction flow → model fine-tuning loop*

From the paper: Mediscript maintains a transcription log specifically for quality assurance and future model fine-tuning, and lets doctors correct transcription output on the fly, with corrections feeding back into the system.

Integration into MediScript AI:

Whenever a user manually corrects a low-confidence or rule-flagged extraction, MediScript AI should log the original OCR/VLM output alongside the corrected value (with user consent, since this is health data) instead of discarding it. Over time this correction log becomes a fine-tuning dataset for the extraction model, turning the safety-driven manual-verification step into a data flywheel that improves accuracy for the handwriting styles and regional languages the system sees most often in practice.

- Persist (original, corrected) pairs from every manual-verification event to a review dataset.
- Periodically fine-tune the extraction model on accumulated corrections, tracked as a versioned dataset.
- Surface correction volume per medicine/language as a model-health metric on the Admin/analytics side.

5. Explicit, Recurring Security Operations

Maps to: *Non-Functional Requirements (Section 10) — Security*

From the paper: Beyond stating that data is encrypted, Mediscript specifies TLS in transit, role-protected databases, auditing and access logs for compliance, and regular security audits and penetration testing as ongoing processes.

Integration into MediScript AI:

MediScript AI's current NFRs state AES-256 at rest and JWT auth but do not specify operational security practices. Adding an explicit process — TLS 1.2+ for all traffic, access logging per record, and a defined cadence for audits and penetration tests — turns security from a one-time architectural claim into a demonstrable, ongoing practice, which is especially relevant given the HIPAA-compliant design goal already stated in the report.

- Add “TLS 1.2+ in transit” and “per-record access logging” as explicit NFRs alongside AES-256 at rest.
- Define an audit/pen-test cadence (e.g., before each major release) as part of the deployment checklist.
- Track access-log review as a recurring Admin Module task, not a one-off setup step.

6. Clean Microservice Separation Between Vision and Extraction

Maps to: *System Architecture (Section 6) — Module 1/2 boundary*

From the paper: Mediscript deploys ASR and NER as two independent microservices with a defined data contract between them: audio in, transcript out; transcript in, structured entities out.

Integration into MediScript AI:

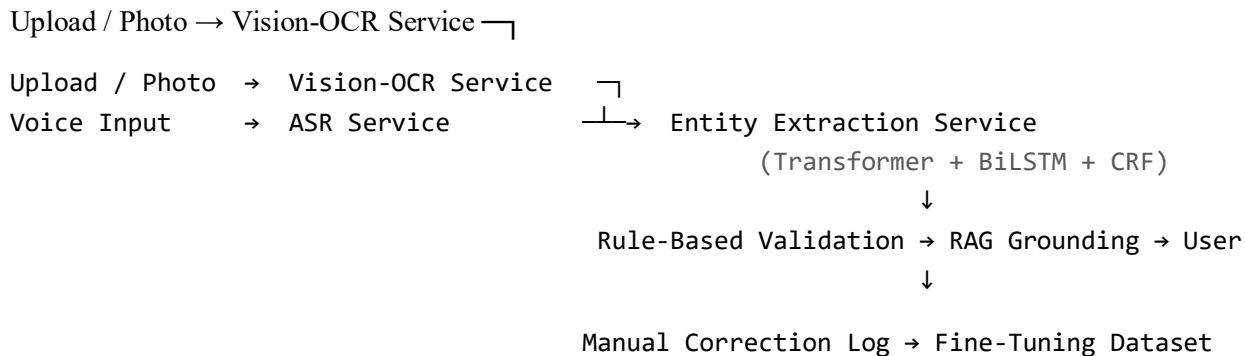
MediScript AI's architecture diagram currently shows OCR + VLM + extraction as a single conceptual flow. Splitting this into two independently deployable services — a Vision/OCR Service (image → raw text) and an Entity Extraction Service (raw text → structured JSON) — with a fixed text-based contract between them mirrors

this proven separation. It also makes the voice input path (Point 2) a natural third producer feeding the same Entity Extraction Service, since both OCR and ASR output plain text into an identical downstream contract.

- Define a single shared “raw prescription text” contract consumed by the extraction service.
- Deploy Vision/OCR and Entity Extraction as separately scalable microservices (matches the FastAPI stack).
- Reuse the same contract for the new voice-to-text path so all three input modes converge on one extractor.

4. Resulting Architectural Adjustment

Applying points 2 and 6 together, the MediScript AI pipeline gains a converged extraction stage: image and PDF uploads pass through the Vision/OCR Service, spoken input passes through a new ASR Service, and both produce plain prescription text consumed by a single Entity Extraction Service built per point 1 (transformer encoder + BiLSTM + CRF). Extracted entities then pass through the rule-based validation layer (point 3) before reaching the user, with any manually corrected fields logged for fine-tuning (point 4), all operating under the recurring security practices defined in point 5.



5. Elements Not Carried Over

Mediscript's video-consultation module (Jitsi Meet integration), appointment scheduling, and doctor-facing dashboard were reviewed but excluded from this integration. MediScript AI is scoped as a patient-facing post-prescription interpretation and assistance tool, not a live telemedicine consultation platform, so these modules fall outside the project's stated objectives and Section 5 scope boundaries.

6. Conclusion

The reference paper and MediScript AI approach the same underlying problem — converting unstructured prescription information into safe, structured data — from opposite ends of the clinical workflow: one at creation time via voice, the other at interpretation time via vision. The six points integrated here (a proven NER architecture, voice as a fallback input, rule-based validation, a correction-driven fine-tuning loop, explicit recurring security operations, and clean service separation) let MediScript AI adopt battle-tested design decisions from a deployed system without changing its core positioning as a RAG-grounded, patient-safety-first prescription assistant.

Source reviewed: Patel, D., Shah, S., Ransing, J., Wani, S., Bendale, B., “*Mediscript: A Low-Cost Multilingual Telemedicine Platform with Voice-Based Prescription Transcription and Automated NER Extraction*,” *International Journal of Research Publication and Reviews*, Vol. 7, Issue 5, May 2026, pp. 11642–11645.